

EXP: 6

6.1 > Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

Program :

```
clc;
clear;
close all;

%% Signal Parameters
fs = 1000; % Sampling Frequency (Hz)
f = 50; % Signal Frequency (Hz)
t = 0:1/fs:0.1; % Time Vector
x = sin(2*pi*f*t); % Original Signal

%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};

% Initialize
xq = cell(1,length(L));
e = cell(1,length(L));
mse = zeros(1,length(L));

%% Original and Quantized Signals
figure;

% Original Signal
subplot(4,1,1);
plot(t,x,'b','LineWidth',1.5);
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Quantization Process
for i = 1:length(L)

% Uniform Quantization
xq{i} = round((x + 1) * (L(i)-1) / 2) * (2/(L(i)-1)) - 1;

% Quantization Error
e{i} = x - xq{i};

% Mean Square Error
mse(i) = mean(e{i}.^2);

% Plot Quantized Signal
subplot(4,1,i+1);
stairs(t,xq{i},colors{i},'LineWidth',1.5);
title(['Quantized Signal ( ' num2str(L(i)) ' Levels)']);
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

end
```

```

%% Display MSE
fprintf('Mean Square Error (MSE)\n');
fprintf('-----\n');
fprintf('8 Levels : %f\n', mse(1));
fprintf('16 Levels : %f\n', mse(2));
fprintf('32 Levels : %f\n', mse(3));

%% Quantization Error Plots
figure;

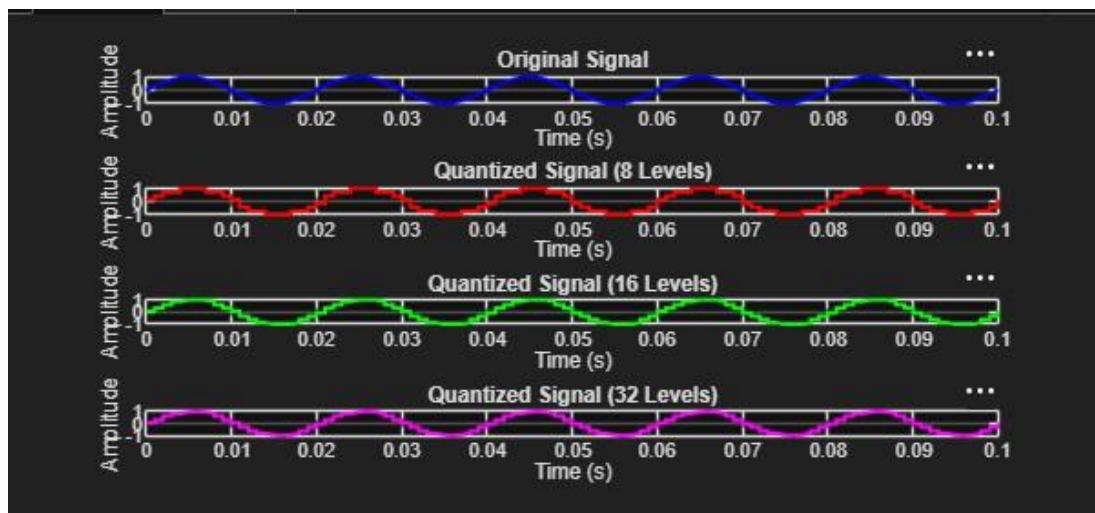
for i = 1:length(L)

subplot(3,1,i);
plot(t,e{i},'LineWidth',1.5);
title(['Quantization Error (' num2str(L(i)) ' Levels)']);
xlabel('Time (s)');
ylabel('Error');
grid on;

end

```

Output





6.2 > Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation Rounding Techniques in MATLAB

Program:

```
clc;
clear;
close all;

%% Generate Original Signal
n = 0:79; % Sample Index
x = sin(2*pi*n/40); % Input Sinusoidal Signal

%% Quantization Factor
q = 8; % Number of Quantization Steps

%% Truncation and Rounding
xt = floor(x*q)/q; % Truncated Signal
xr = round(x*q)/q; % Rounded Signal

%% Quantization Errors
et = x - xt; % Truncation Error
er = x - xr; % Rounding Error

%% Mean Square Error
mse_t = mean(et.^2);
mse_r = mean(er.^2);

%% Figure 1: Original vs Rounded Signal
figure;
plot(n,x,'b','LineWidth',1.5);
hold on;
plot(n,xr,'r','LineWidth',1.5);
grid on;
xlabel('Sample Index');
ylabel('Amplitude');
title('Original vs Rounded Quantized Signal');
legend('Original','Rounded');

%% Figure 2: Truncation vs Rounding
figure;
```

```

plot(n,x,'b','LineWidth',1.5);
hold on;
plot(n,xt,'r','LineWidth',1.5);
plot(n,xr,'g','LineWidth',1.5);
grid on;
xlabel('Sample Index');
ylabel('Amplitude');
title('Truncation vs Rounding');
legend('Original','Truncated','Rounded');

%% Figure 3: Quantization Error
figure;
stem(n,et,'r','filled');
hold on;
stem(n,er,'g','filled');
grid on;
xlabel('Sample Index');
ylabel('Error');
title('Quantization Error');
legend('Truncation Error','Rounding Error');

%% Figure 4: MSE Comparison
figure;
bar([mse_t mse_r]);
set(gca,'XTickLabel',{'Truncation','Rounding'});
ylabel('Mean Square Error');
title('MSE Comparison');
grid on;

%% Display Results
fprintf('Mean Square Error (Truncation) = %.6f\n', mse_t);
fprintf('Mean Square Error (Rounding) = %.6f\n', mse_r);

```

Output:

